Applied Machine Learning

Learning goals

- Basics of mlr3tuning
- Parameter Transformations
- Nested Resampling in mlr3

ADVANCED MACHINE LEARNING

Recap: Tuning

Intro to mlr3tuning

Parameter Transformation

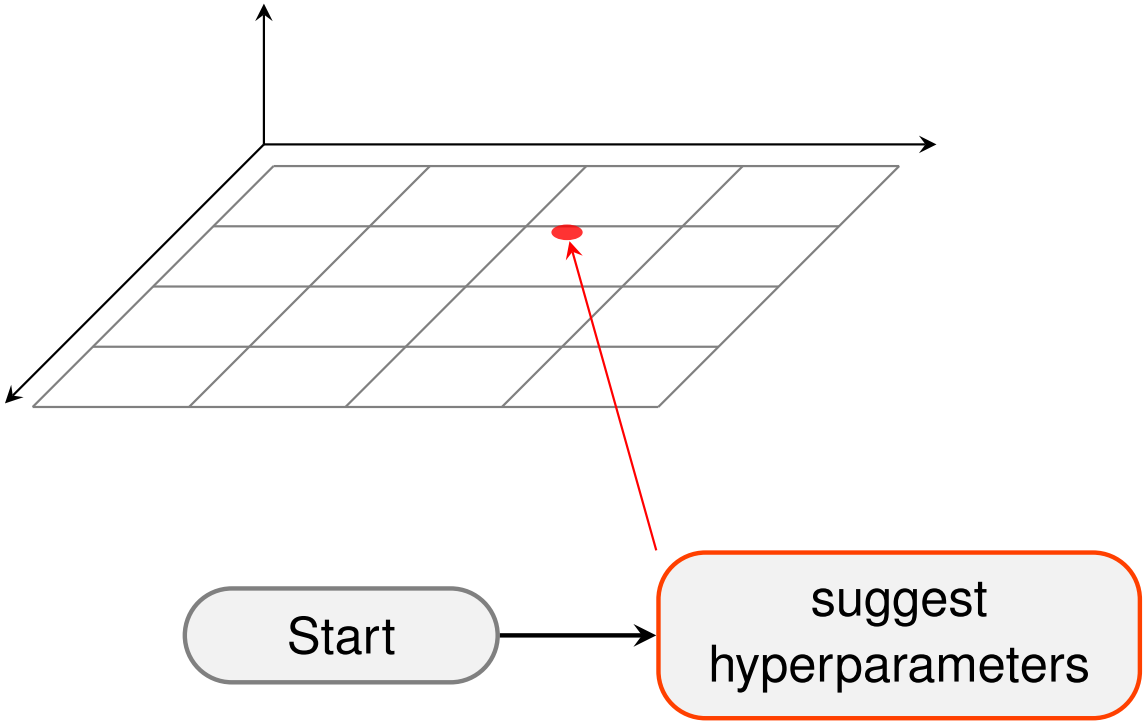
AutoTuner and Nested Resampling in mlr3

TUNING

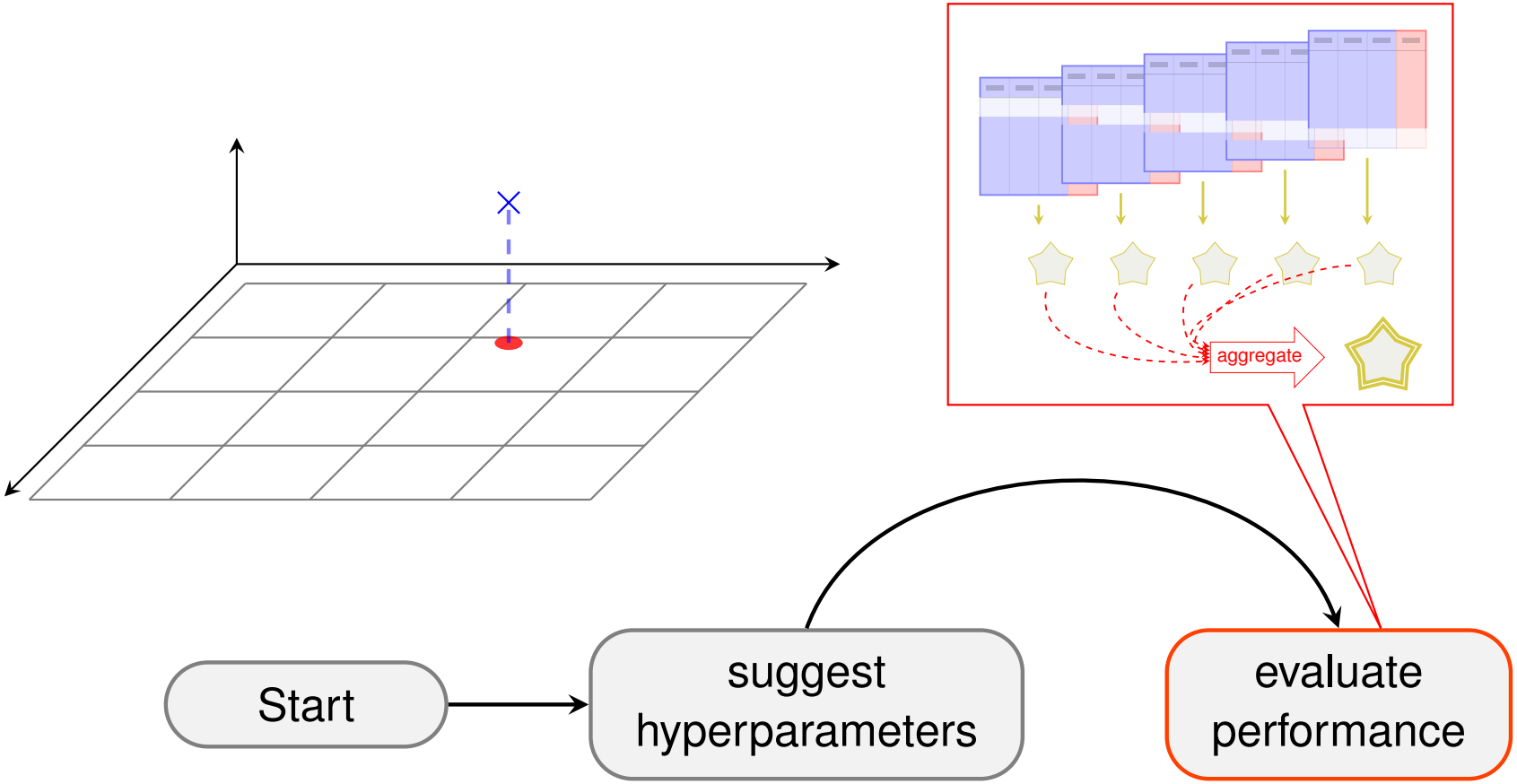
- Behavior of most methods depends on *hyperparameters*
- We want to choose them so our algorithm performs well
- Good hyperparameters are data-dependent

$\Rightarrow$ We do *black box optimization* ("Try stuff and see what works")

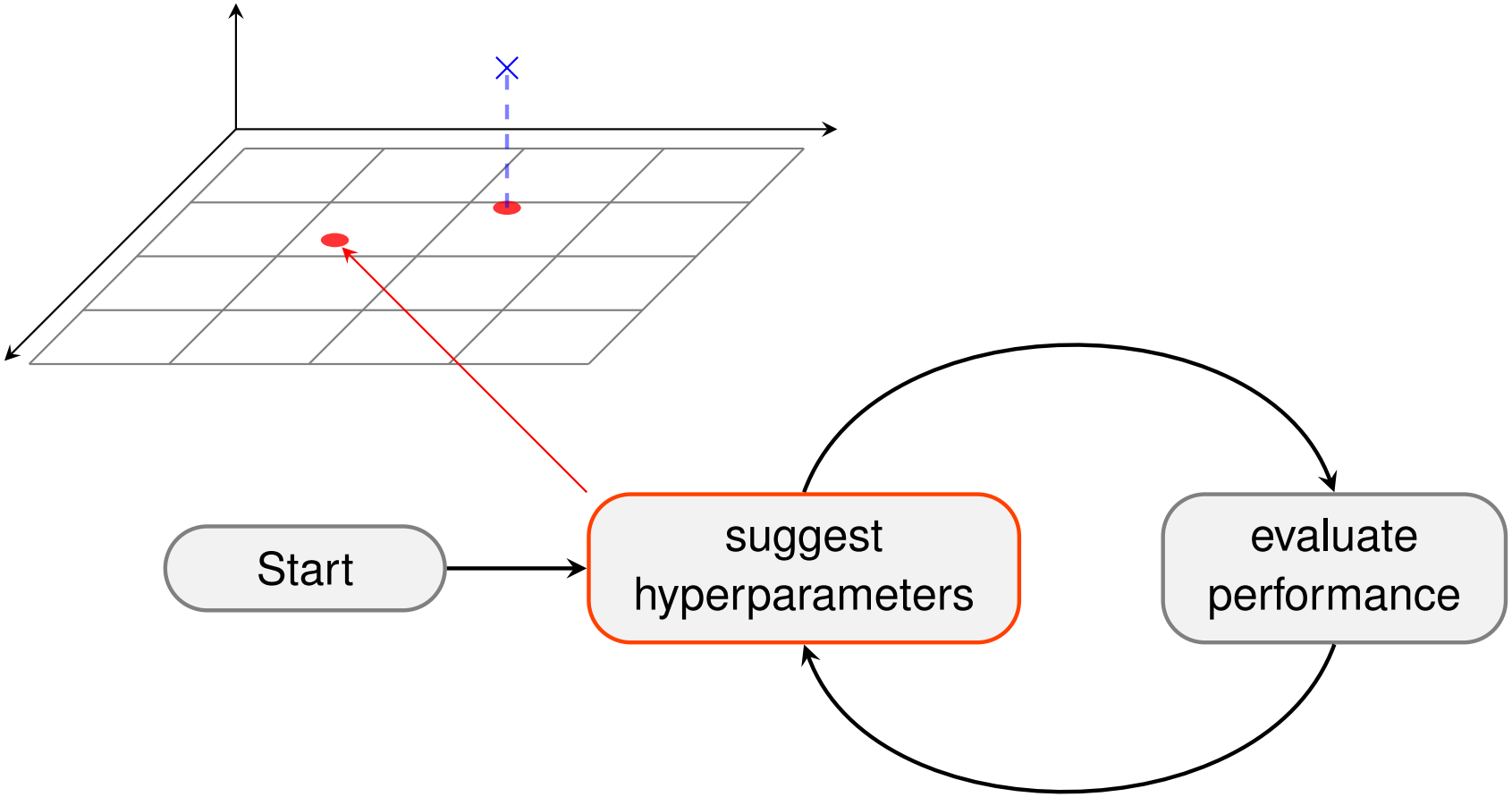
TUNING



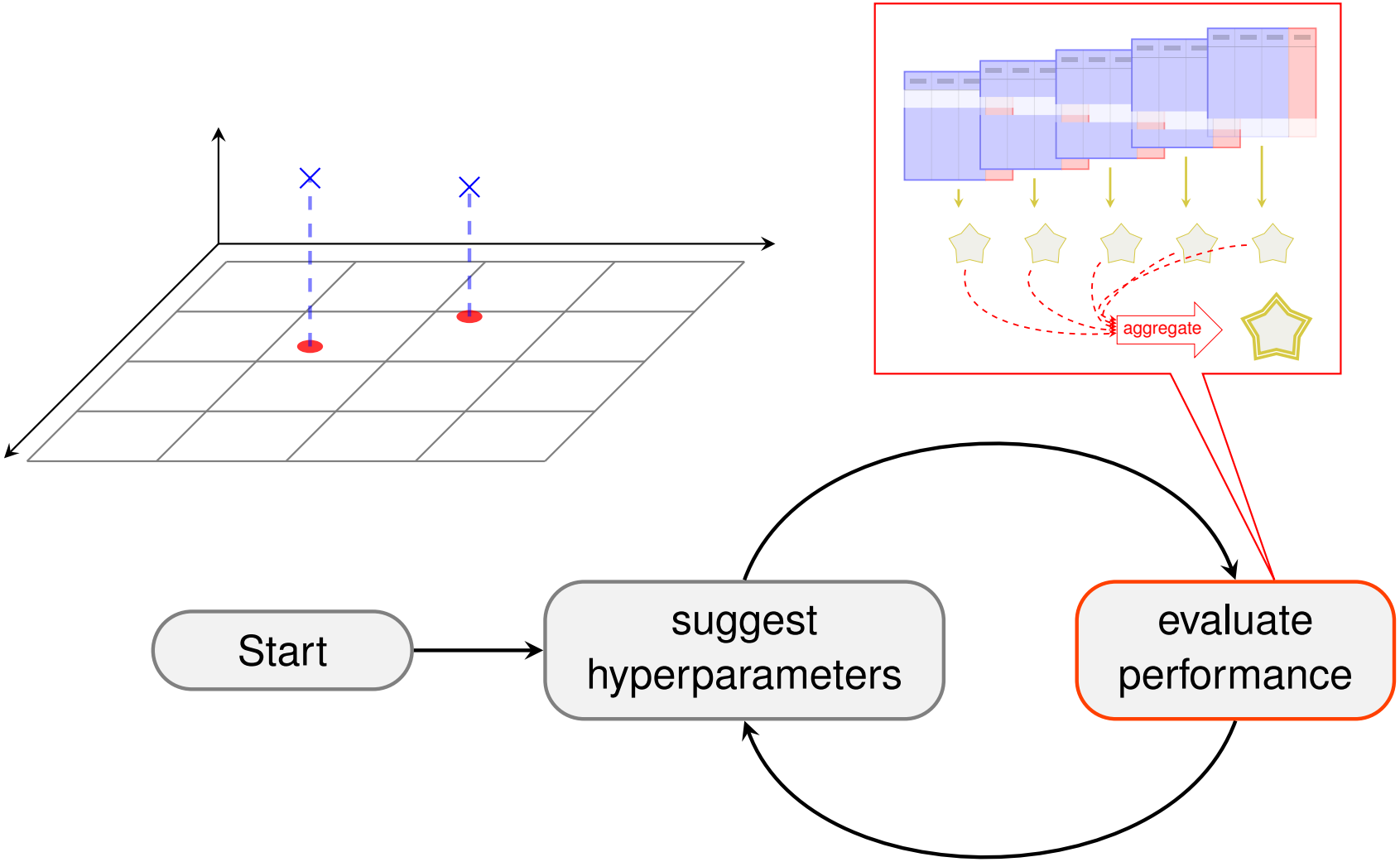
TUNING


aggregate
Start
suggest hyperparameters
evaluate performance

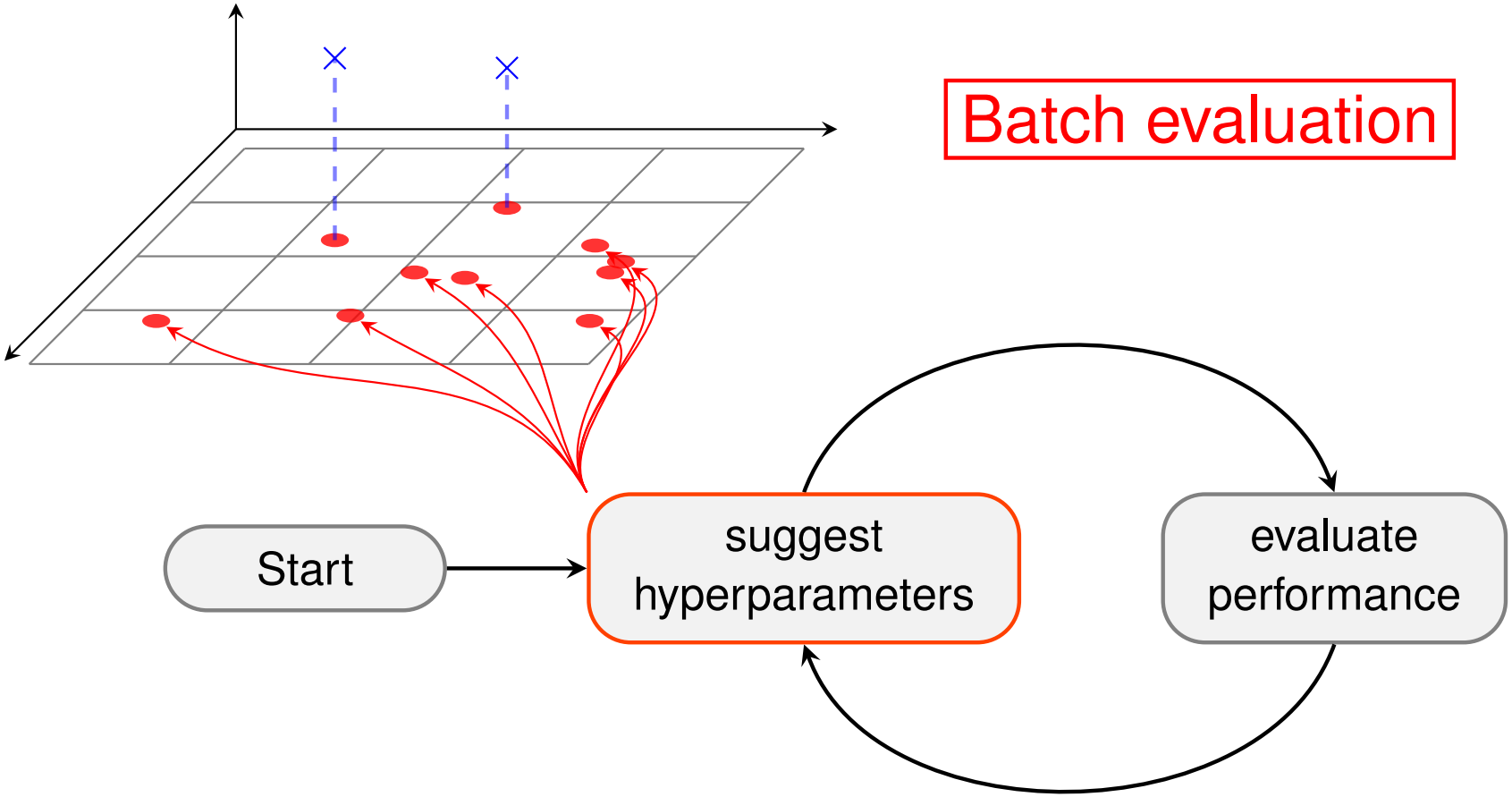
TUNING



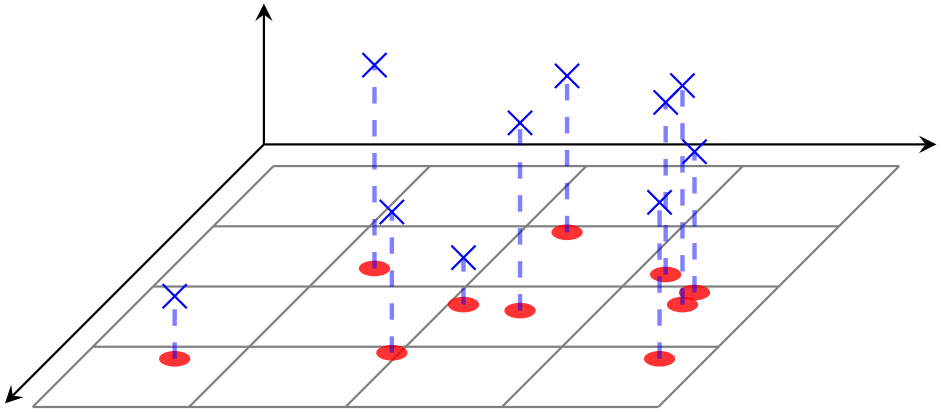
TUNING

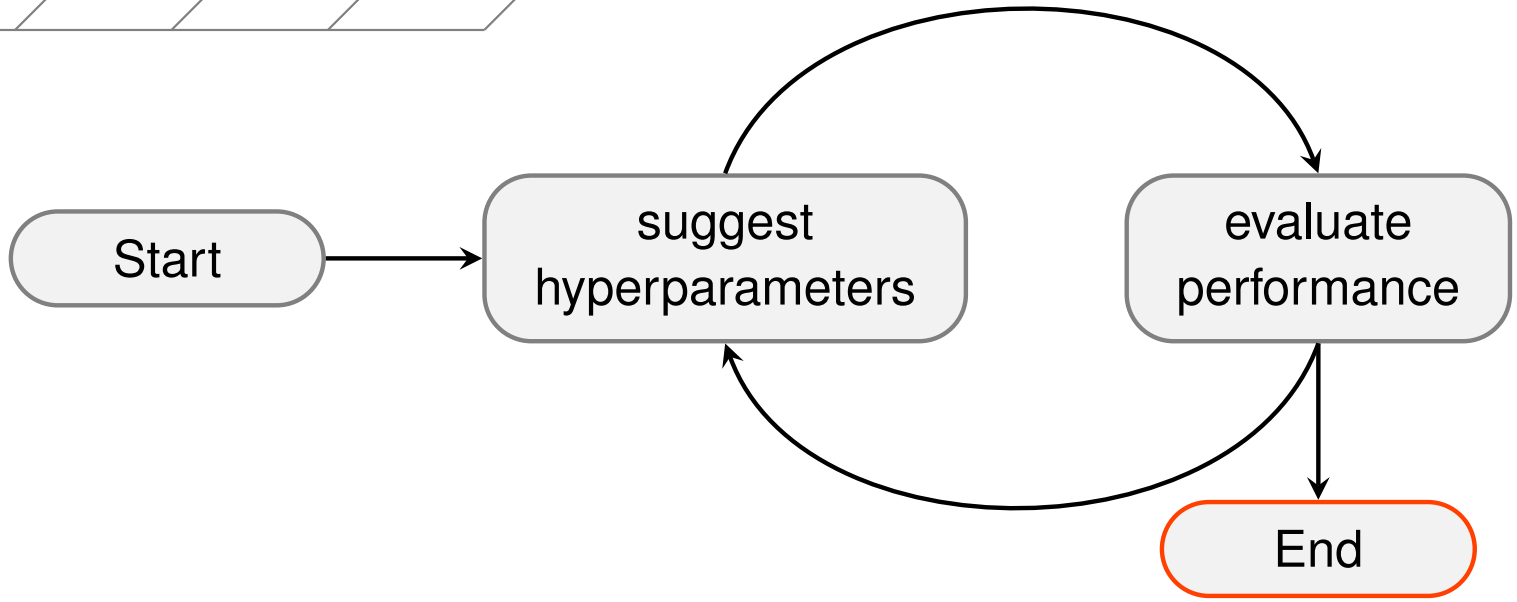

aggregate
Start
suggest hyperparameters
evaluate performance

TUNING


Batch evaluation
Start
suggest hyperparameters
evaluate performance

TUNING





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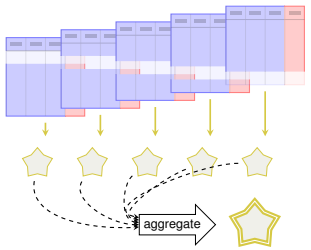
- We want to choose them so our algorithm performs well
- Good hyperparameters are data-dependent

$\Rightarrow$ We do *black box optimization* ("Try stuff and see what works")

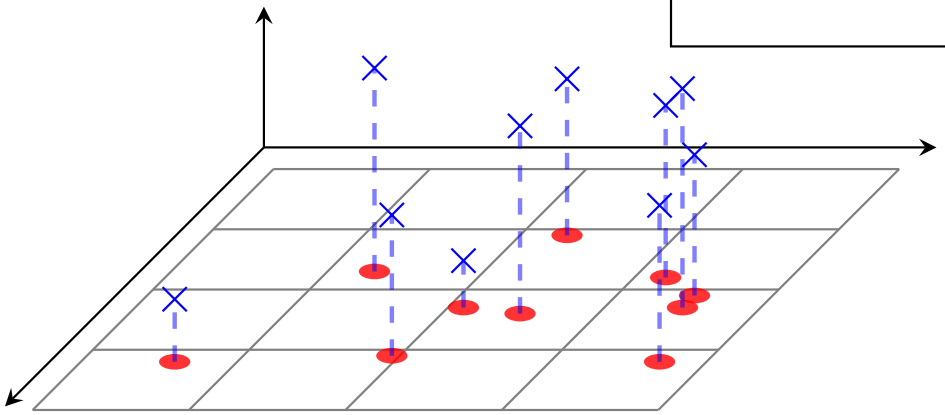
Tuning toolbox for `mlr3`:

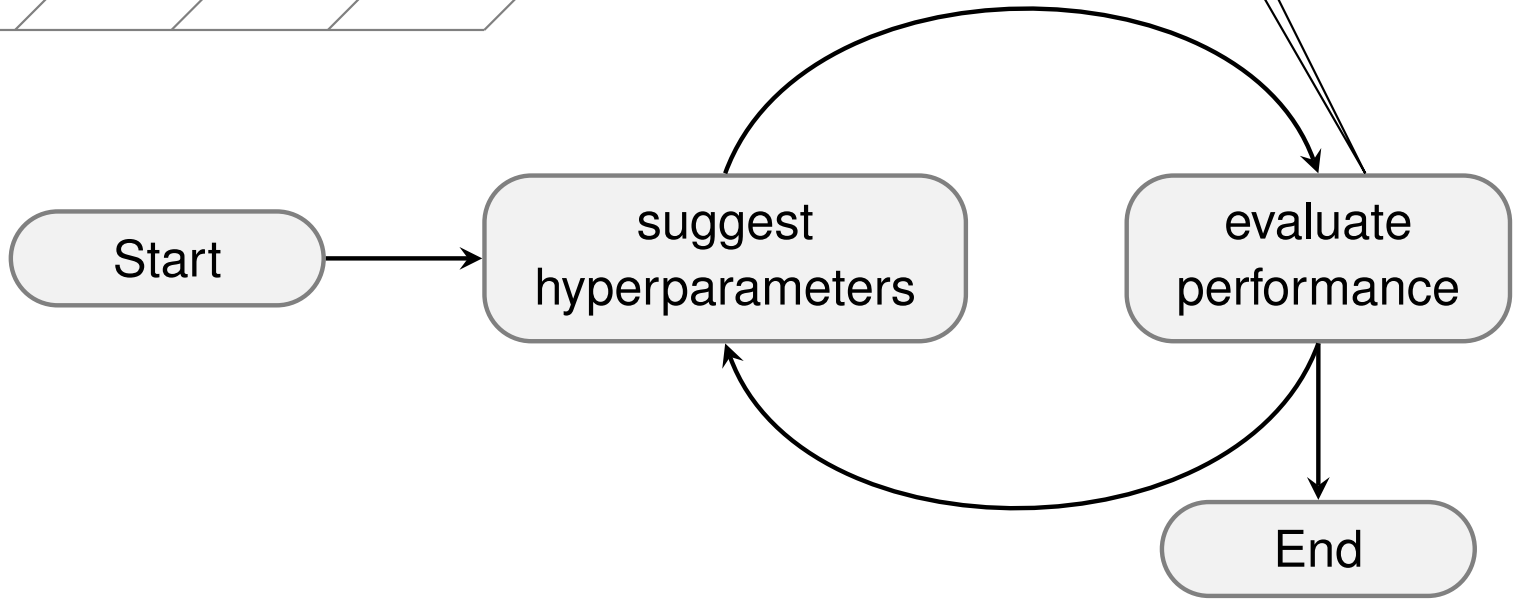
```
library("bbotk")
library("mlr3tuning")
```

TUNING

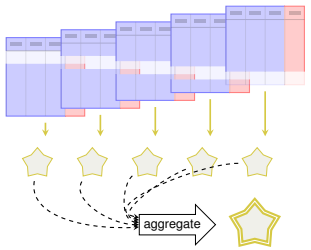


`Learner(lrn())`
`Task(as_task_*())`
`Resampling(rsmp())`
`Measure(msr())`

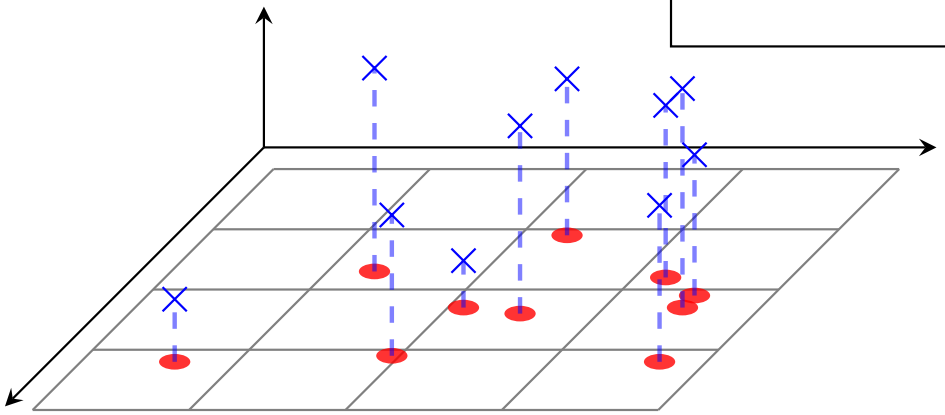




TUNING



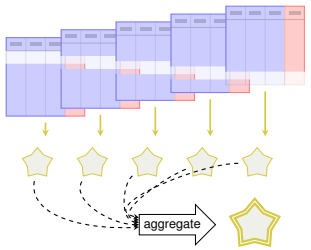
`Learner` (`lrn()`)
`Task` (`as_task_*()`)
`Resampling` (`rsmp()`)
`Measure` (`msr()`)



Start → suggest hyperparameters → evaluate performance → End

search space
`Tuner`

TUNING



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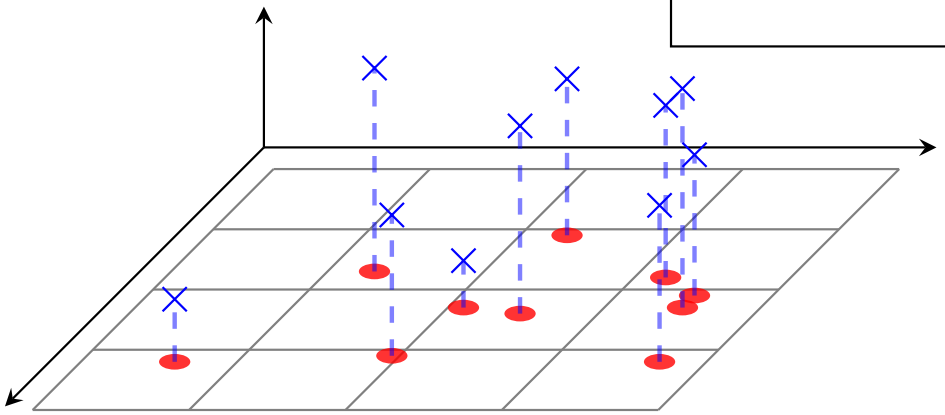
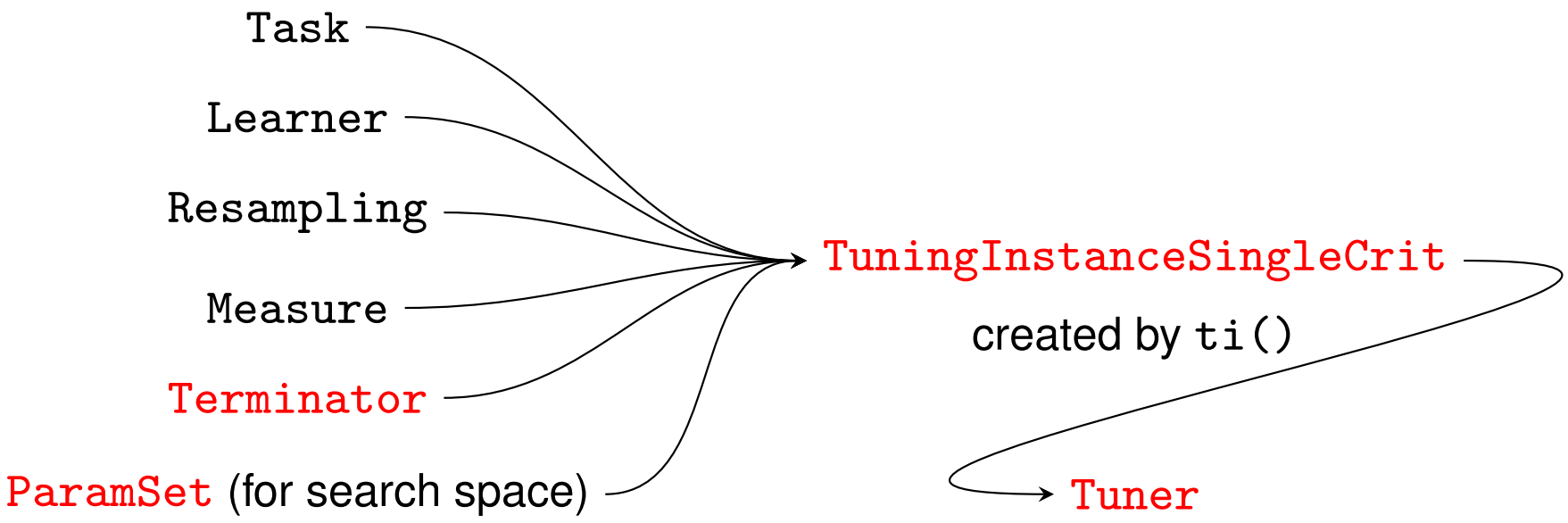


Start → suggest hyperparameters → evaluate performance → End

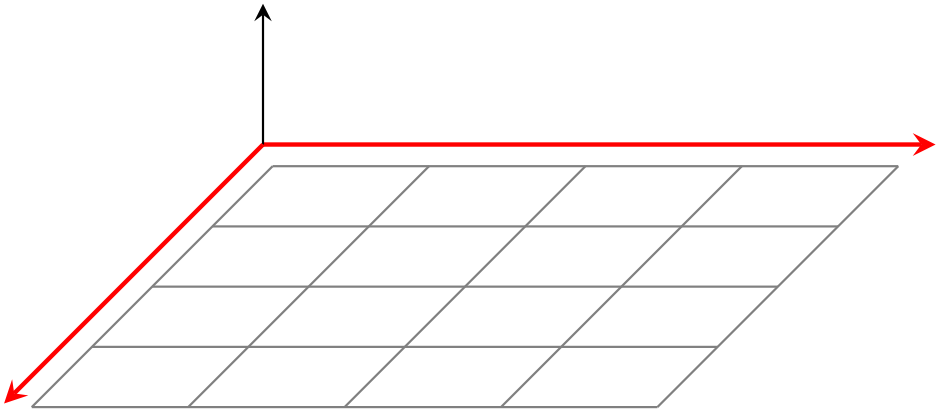
search space
`Tuner`

`Terminator` (when to finish)

OBJECTS IN TUNING



SEARCH SPACE SHORT FORM



```r
ps(id1 = domain1, id2 = domain2, ...)
```

Numerical parameter	`p_dbl(lower, upper)`
Integer parameter	`p_int(lower, upper)`
Discrete parameter	`p_fct(levels)`
Logical parameter	`p_lgl()`
Untyped parameter	`p_uty()`

```r
library("paradox")
searchspace_knn = ps(
  "k" = p_int(lower = 1, upper = 35)
)
```

SEARCH SPACE IN LEARNER DEFINITION

Specify search space for the hyperparameter to be tuned directly when creating the learner using `to_tune`:

```
library("paradox")
learner = lrn("classif.kknn", k = to_tune(lower = 1, upper = 35))
learner$param_set

#> <ParamSet>
#>             id    class lower upper nlevels default                value
#>         <char>   <char> <num> <num>   <num>  <list>               <list>
#> 1:           k ParamInt     1   Inf     Inf       7 <RangeTuneToken[2]>
#> 2:    distance ParamDbl     0   Inf     Inf       2
#> 3:      kernel ParamFct    NA    NA      10 optimal
#> 4:       scale ParamLgl    NA    NA       2    TRUE
#> 5:     ykernel ParamUty    NA    NA     Inf
#> 6: store_model ParamLgl    NA    NA       2   FALSE
```

Note: Calling `$train()` on a learner with such a specified search space does not work.

TERMINATION

- Tuning needs a *termination condition*: when to finish
- `mlr_terminators` dictionary:
-
```
as.data.table(mlr_terminators)[, c("key", "label", "properties")]
#> Key: <key>
#>                 key                label             properties
#>              <char>               <char>                 <list>
#> 1:       clock_time           Clock Time single-crit,multi-crit
#> 2:            combo          Combination single-crit,multi-crit
#> 3:            evals Number of Evaluation single-crit,multi-crit
#> 4:             none                 None single-crit,multi-crit
#> 5:     perf_reached    Performance Level            single-crit
#> 6:         run_time             Run Time single-crit,multi-crit
#> 7:       stagnation           Stagnation            single-crit
#> 8: stagnation_batch     Stagnation Batch            single-crit
```

TERMINATION

- Tuning needs a *termination condition*: when to finish
- `mlr_terminators` dictionary:
-
```
as.data.table(mlr_terminators)[, c("key", "label", "properties")]
#> Key: <key>
#>                 key                label             properties
#>              <char>               <char>                 <list>
#> 1:       clock_time           Clock Time single-crit,multi-crit
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#> 6:         run_time             Run Time single-crit,multi-crit
#> 7:       stagnation           Stagnation            single-crit
#> 8: stagnation_batch     Stagnation Batch            single-crit
```
- Construct `Terminator` via `trm()` short form:
-
```
trm("evals", n_evals = 20)
#> <TerminatorEvals>: Number of Evaluation
#> * Parameters: n_evals=20, k=0
```

TUNING METHOD

- Choose a *tuning method* via `mlr_tuners` dictionary:

```
as.data.table(mlr_tuners)[-1, c("key", "label")]
```

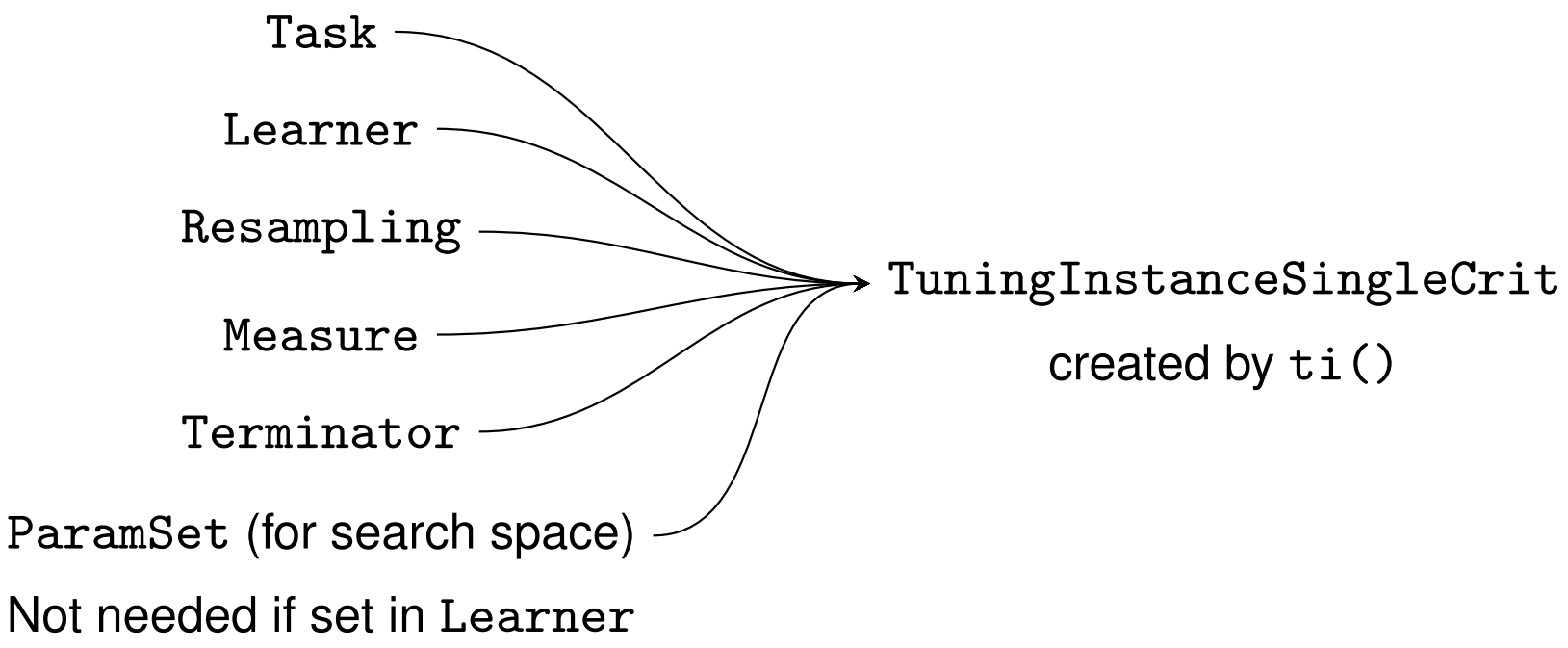
```
#> Key: <key>
#>                   key                            label
#>                <char>                           <char>
#> 1:      design_points                     Design Points
#> 2:              gensa Generalized Simulated Annealing
#> 3:        grid_search                       Grid Search
#> 4:          hyperband                         Hyperband
#> 5:              irace                    Iterated Racing
#> 6:                mbo          Model Based Optimization
#> 7:             nloptr           Non-linear Optimization
#> 8:      random_search                     Random Search
#> 9: successive_halving                Successive Halving
```

- Construct `Tuner` via `tnr()` (`batch_size` for parallelization):

```
gsearch = tnr("grid_search", resolution = 7)
print(gsearch)
```

```
#> <TunerGridSearch>: Grid Search
#> * Parameters: resolution=7, batch_size=1
#> * Parameter classes: ParamLgl, ParamInt, ParamDbl, ParamFct
#> * Properties: dependencies, single-crit, multi-crit
#> * Packages: mlr3tuning
```

CALLING THE TUNER




```
inst = ti(task = tsk("german_credit"),
  learner = lrn("classif.kknn"),
  resampling = rsmp("holdout"),
  measure = msr("classif.ce"),
  terminator = trm("evals", n_evals = 2),
  search_space = searchspace_knn
)
```

CALLING THE TUNER

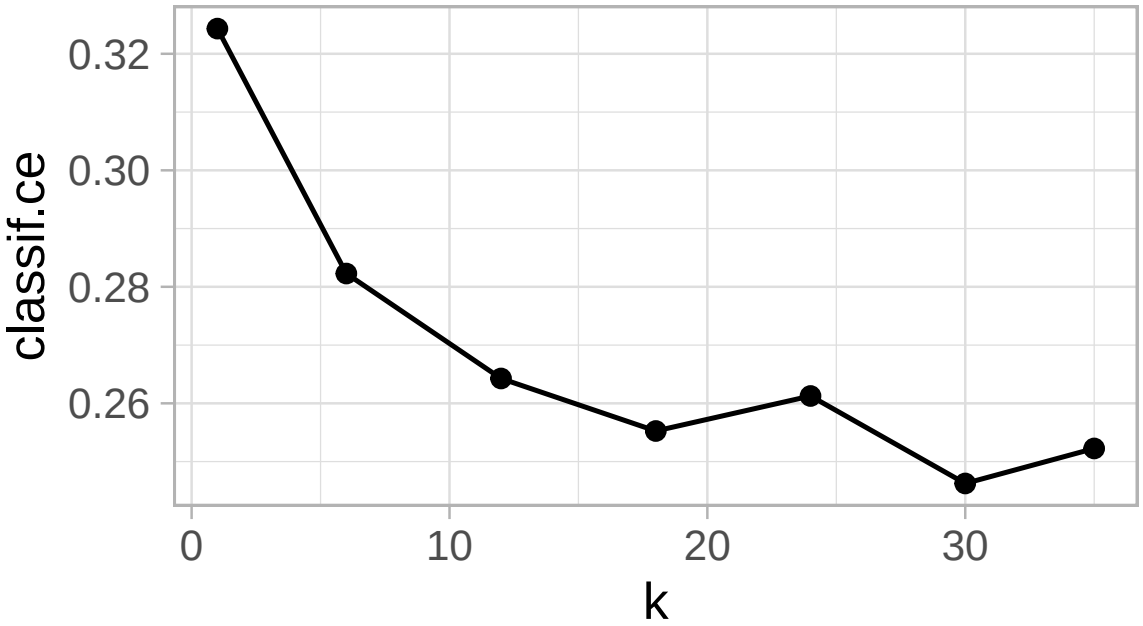
```
gsearch$optimize(inst)
#> INFO  [07:05:21.749] [bbotk] Starting to optimize 1 parameter(s) with '<TunerGridSearch
#> INFO  [07:05:21.846] [bbotk] Evaluating 1 configuration(s)
#> INFO  [07:05:24.370] [bbotk] Result of batch 1:
#> INFO  [07:05:24.377] [bbotk]   k classif.ce warnings errors runtime_learners
#> INFO  [07:05:24.377] [bbotk]  12       0.26        0      0             0.25
#> INFO  [07:05:24.377] [bbotk]                                uhash
#> INFO  [07:05:24.377] [bbotk]  50fda423-4a96-4c6a-ba8f-dc58fb439a73
#> INFO  [07:05:24.381] [bbotk] Evaluating 1 configuration(s)
#> INFO  [07:05:24.644] [bbotk] Result of batch 2:
#> INFO  [07:05:24.651] [bbotk]  k classif.ce warnings errors runtime_learners
#> INFO  [07:05:24.651] [bbotk]  1       0.32        0      0             0.13
#> INFO  [07:05:24.651] [bbotk]                                uhash
#> INFO  [07:05:24.651] [bbotk]  e0d8552b-bfc4-4419-855d-0bbbe4414e4c
#> INFO  [07:05:24.665] [bbotk] Finished optimizing after 2 evaluation(s)
#> INFO  [07:05:24.667] [bbotk] Result:
#> INFO  [07:05:24.672] [bbotk]      k learner_param_vals  x_domain classif.ce
#> INFO  [07:05:24.672] [bbotk]  <int>             <list>    <list>      <num>
#> INFO  [07:05:24.672] [bbotk]     12          <list[1]> <list[1]>       0.26
#>        k learner_param_vals  x_domain classif.ce
#>    <int>             <list>    <list>      <num>
#> 1:    12          <list[1]> <list[1]>       0.26
```

TUNING RESULTS

```
inst = ti(task = tsk("german_credit"), learner = lrn("classif.kknn"),
  resampling = rsmp("holdout"), measure = msr("classif.ce"),
  terminator = trm("none"), search_space = searchspace_knn)
gsearch = tnr("grid_search", resolution = 7)
gsearch$optimize(inst)

#>        k learner_param_vals  x_domain classif.ce
#>    <int>             <list>    <list>      <num>
#> 1:    30          <list[1]> <list[1]>       0.25

inst$archive # results are stored in tuning instance

#> <ArchiveTuning> with 7 evaluations
#>        k classif.ce batch_nr warnings errors
#>    <int>      <num>    <int>    <int>  <int>
#> 1:     6       0.28        1        0      0
#> 2:    24       0.26        2        0      0
#> 3:    18       0.26        3        0      0
#> 4:    30       0.25        4        0      0
#> 5:    12       0.26        5        0      0
#> 6:    35       0.25        6        0      0
#> 7:     1       0.32        7        0      0
```

TUNING RESULTS - VISUALIZATION

```
ggplot(as.data.table(inst$archive), aes(x = k, y = classif.ce)) +
  geom_line() + geom_point()
```



RECAP

1. Create a `Task`, `Learner`, `Resampling`, `Measure`, `Terminator` (defines when to stop), and a `ParamSet` (defines the search space):

```
task = tsk("german_credit")
learner = lrn("classif.kknn")
resampling = rsmp("holdout")
measure = msr("classif.ce")
terminator = trm("none")
searchspace_knn =  ps("k" = p_int(lower = 1, upper = 35))
```

2. Create the `TuningInstanceSingleCrit` object:

```
inst = ti(task, learner, resampling, measure,
  terminator, searchspace_knn)
```

3. Create the `Tuner` (tuning method) and optimize the learner by passing over the previously created instance to the `$optimize` method:

```
gsearch = tnr("grid_search", resolution = 7)
gsearch$optimize(inst)
#>        k learner_param_vals  x_domain classif.ce
#>    <int>             <list>    <list>      <num>
#> 1:    24          <list[1]> <list[1]>       0.24
```

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PARAMETER TRANSFORMATION

- Sometimes we do not want to optimize over an evenly spaced range
- $k = 1$ vs. $k = 2$ probably more interesting than $k = 101$ vs. $k = 102$

$\Rightarrow$ Use transformations (part of `ParamSet`)

PARAMETER TRANSFORMATION

- Sometimes we do not want to optimize over an evenly spaced range
- $k = 1$ vs. $k = 2$ probably more interesting than $k = 101$ vs. $k = 102$

$\Rightarrow$ Use transformations (part of `ParamSet`)

Example:

1. optimize from $\log(1) \ldots \log(100)$
2. transform by $\exp()$ in `trafo` function
3. don't forget to `round` (k must be integer)

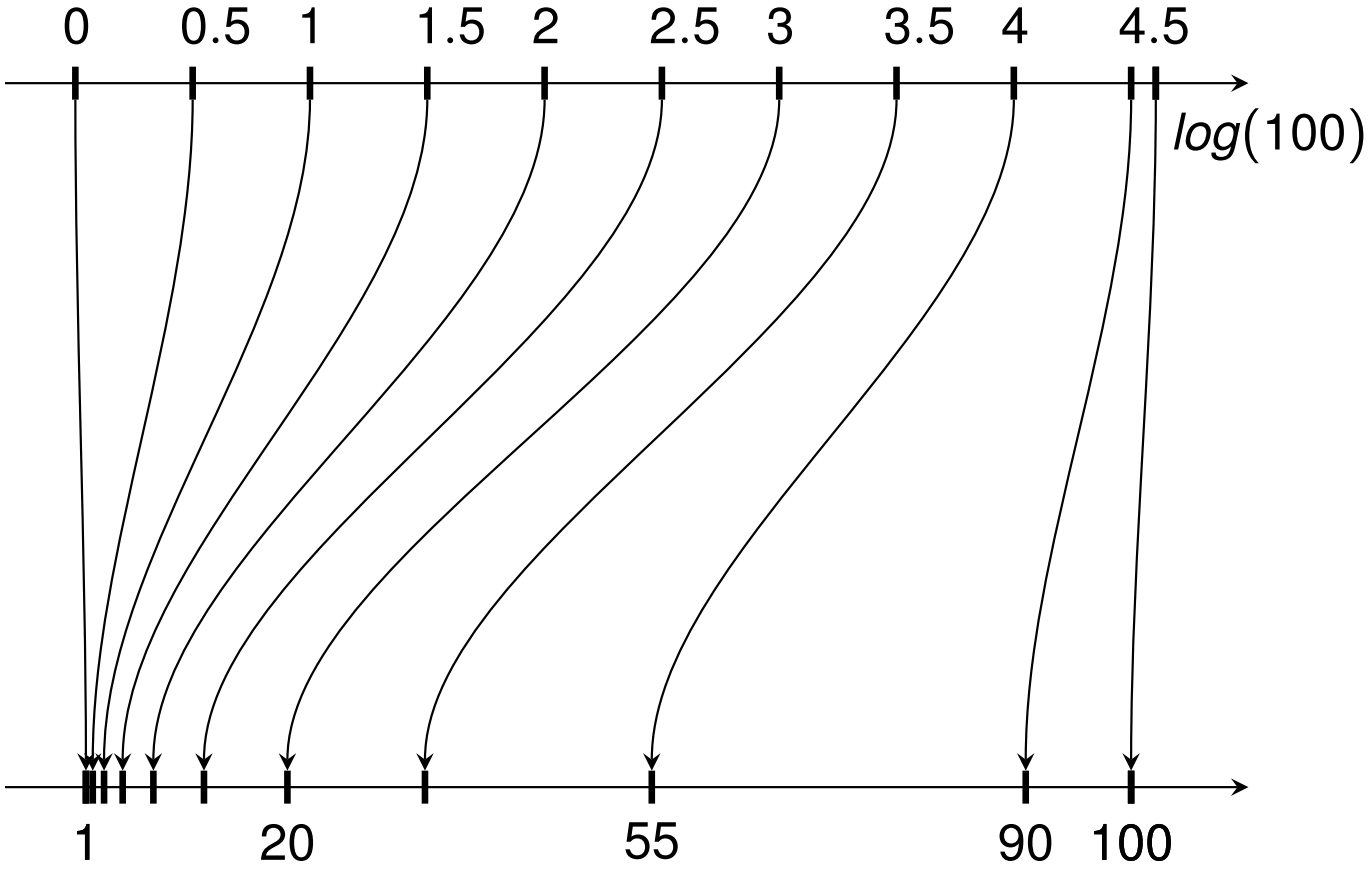
```
searchspace_knn_trafo = ps(
  "k" = p_dbl(log(1), log(35), trafo = function(x) round(exp(x)))
)
```

PARAMETER TRANSFORMATION

What is our transformation doing?

PARAMETER TRANSFORMATION

What is our transformation doing?



PARAMETER TRANSFORMATION

Tune again with `searchspace_knn_trafo` ...

```
inst_trafo = ti(task, learner, resampling, measure,
    terminator, searchspace_knn_trafo)
gsearch$optimize(inst_trafo)

#>        k learner_param_vals  x_domain classif.ce
#>    <num>             <list>    <list>      <num>
#> 1:   3.6          <list[1]> <list[1]>       0.21
```

PARAMETER TRANSFORMATION

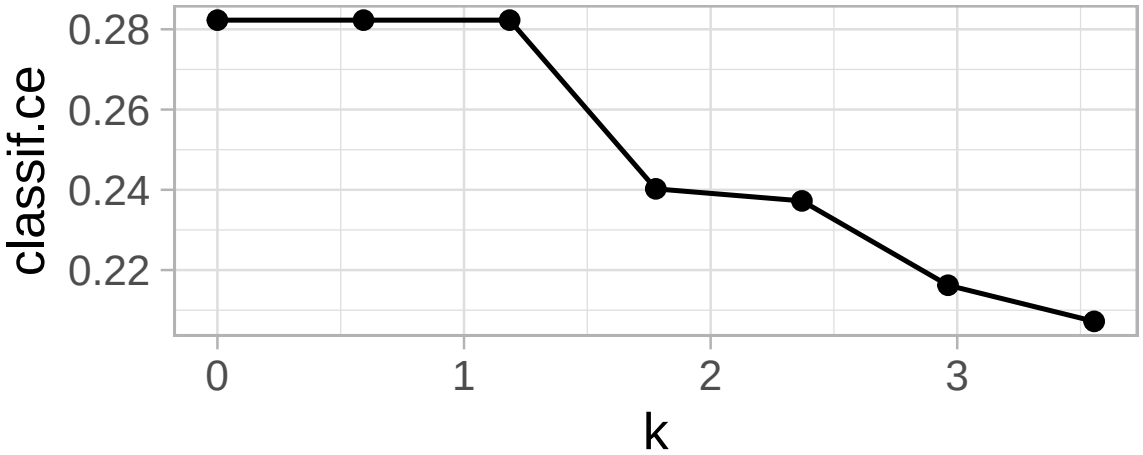
Tune again with `searchspace_knn_trafo` ...

```
inst_trafo = ti(task, learner, resampling, measure,
    terminator, searchspace_knn_trafo)
gsearch$optimize(inst_trafo)

#>        k learner_param_vals  x_domain classif.ce
#>    <num>             <list>    <list>      <num>
#> 1:   3.6          <list[1]> <list[1]>       0.21
```

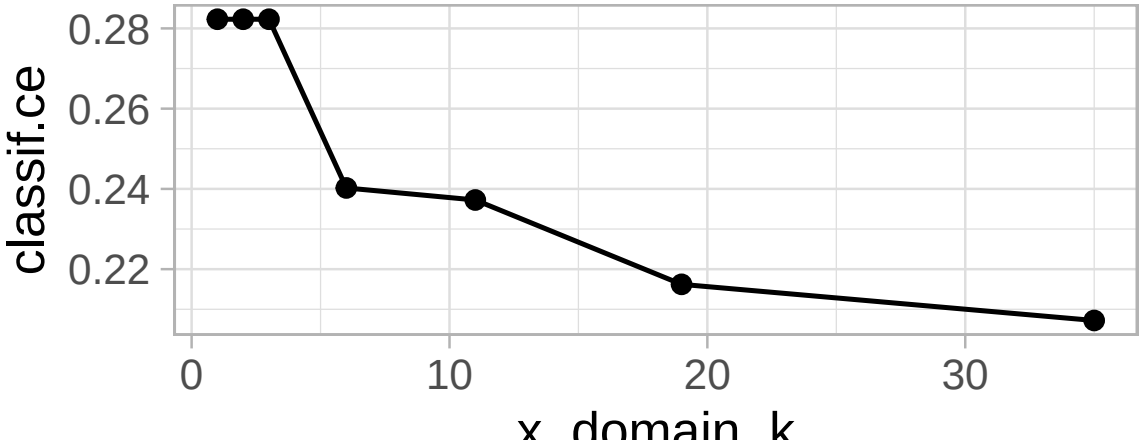
```
arv_trafo = as.data.table(inst_trafo$archive)
arv_trafo[, 1:4]

#>        k classif.ce x_domain_k runtime_learners
#>    <num>      <num>      <num>            <num>
#> 1:  2.96       0.22         19             0.13
#> 2:  3.56       0.21         35             0.19
#> 3:  1.78       0.24          6             0.11
#> 4:  1.19       0.28          3             0.17
#> 5:  0.59       0.28          2             0.11
#> 6:  0.00       0.28          1             0.12
#> 7:  2.37       0.24         11             0.12
```

PARAMETER TRANSFORMATION

```
ggplot(data = arv_trafo, aes(x = k, y = classif.ce)) +
  geom_line() + geom_point()
```




```
ggplot(data = arv_trafo, aes(x = x_domain_k, y = classif.ce)) +
  geom_line() + geom_point()
```



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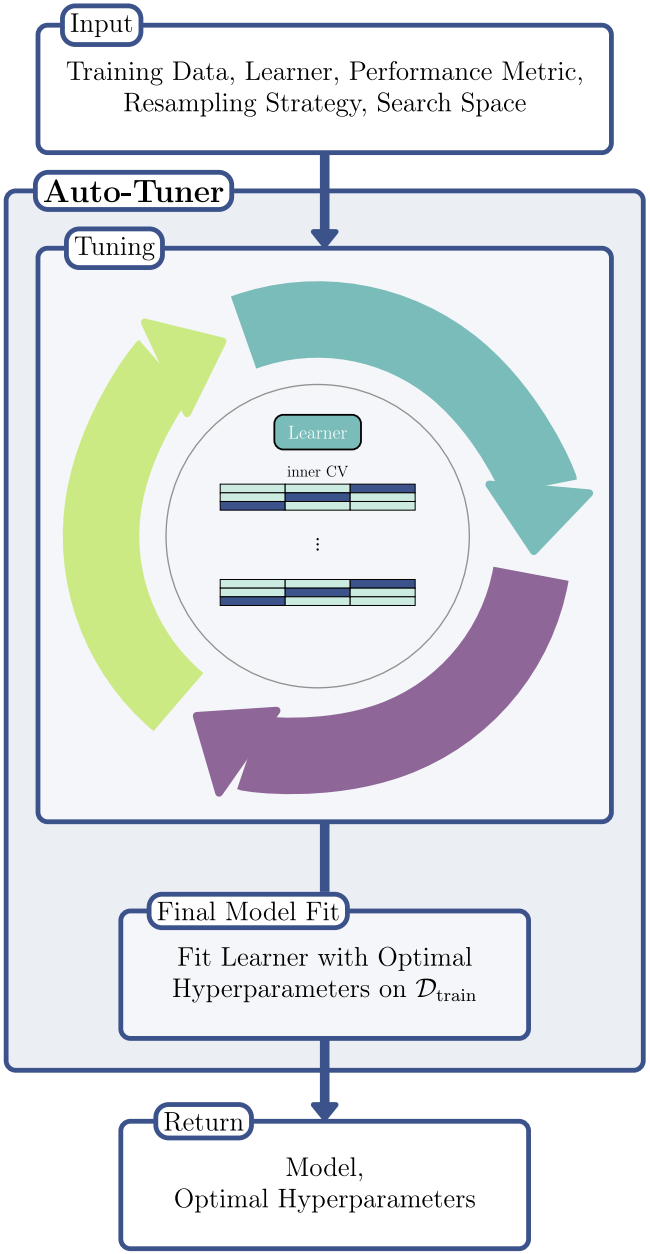
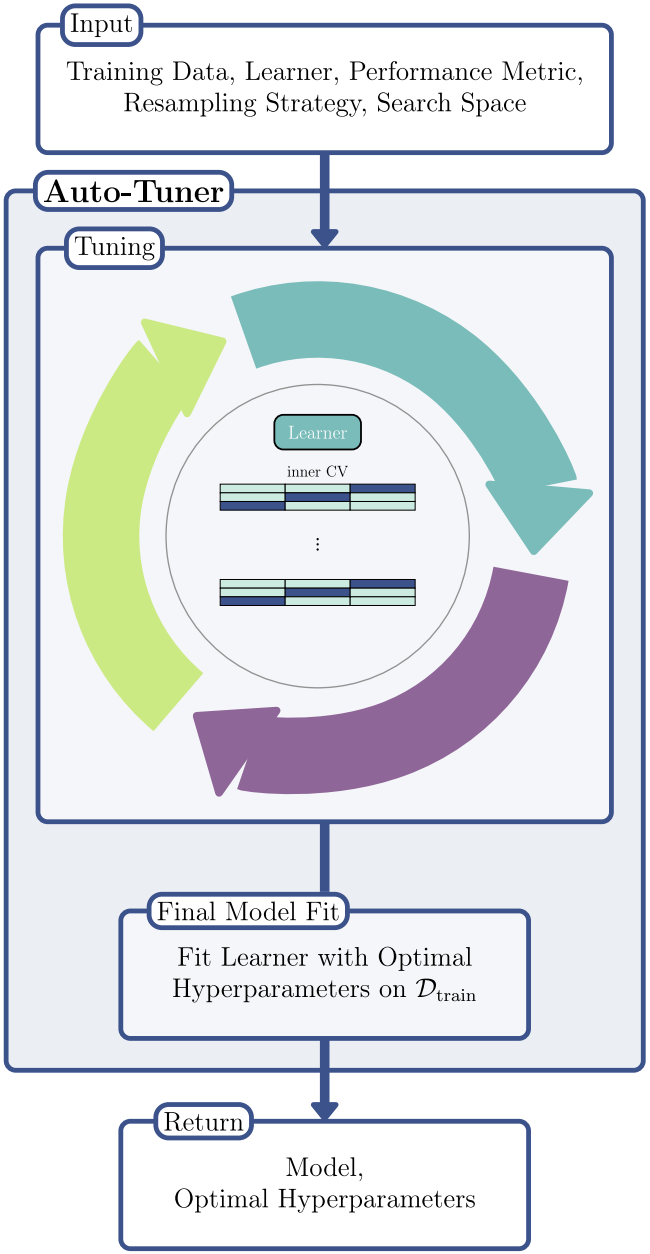


Treat tuning as part of the learning process!

- Training:
  1. Tune model using (inner) resampling
  2. Train final model with optimal parameters on all data
- Predicting: Use final model

AUTOTUNER



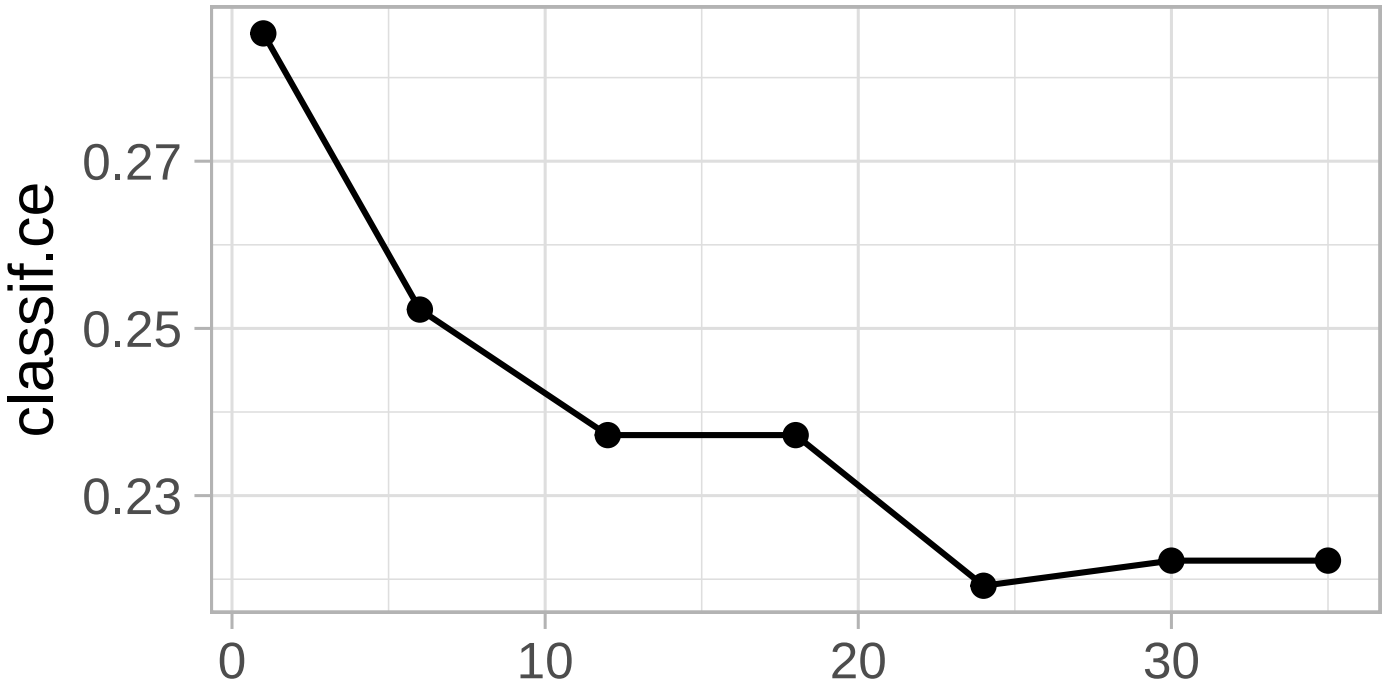
Treat tuning as part of the learning process!

- Training:
  1. Tune model using (inner) resampling
  2. Train final model with optimal parameters on all data
- Predicting: Use final model

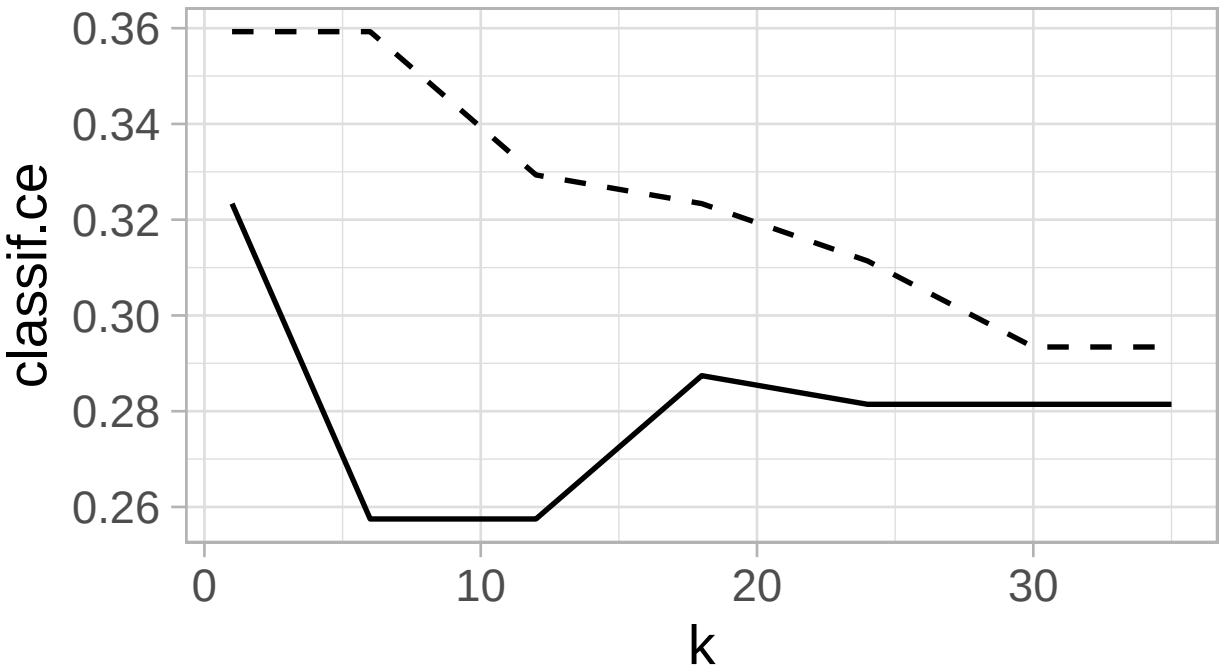
```
optlrn = auto_tuner(
  tuner = tnr("grid_search", resolution = 7),
  learner = lrn("classif.kknn", k = to_tune(1, 35)),
  resampling = rsmp("holdout"),
  measure = msr("classif.ce"),
  terminator = trm("none"))
optlrn$train(tsk("german_credit"))
optlrn$learner # learner with best found HP

#> <LearnerClassifKKNN:classif.kknn>: k-Nearest-Neighbor
#> * Model: list
#> * Parameters: k=24
#> * Packages: mlr3, mlr3learners, kknn
#> * Predict Types:  [response], prob
#> * Feature Types: logical, integer, numeric,
#>   factor, ordered
#> * Properties: multiclass, twoclass
```

AUTOTUNER - ANALYZE TUNING RESULTS

```
arv = as.data.table(optlrn$tuning_instance$archive)
ggplot(arv, aes(x = k, y = classif.ce)) +
  geom_line() + geom_point() + xlab("")
```



AUTOTUNER - NESTED RESAMPLING

To estimate **tuned learner** performance, an outer resampling is required (performs nested resampling):

```
rr = resample(task = tsk("german_credit"), learner = optlrn,
  resampling = rsmp("cv", folds = 2), store_models = TRUE)
arv1 = as.data.table(rr$learners[[1]]$tuning_instance$archive)
arv2 = as.data.table(rr$learners[[2]]$tuning_instance$archive)
ggplot() +
  geom_line(data = arv1, aes(x = k, y = classif.ce)) +
  geom_line(data = arv2, aes(x = k, y = classif.ce), linetype = 2)
```